

Embedded Safety-Aligned Intelligence for Multi-Agent Reinforcement Learning

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Abstract

We introduce *Embedded Safety-Aligned Intelligence* (ESAI), a theoretical framework for multi-agent reinforcement learning (MARL) that embeds alignment constraints directly into agents’ internal representations via differentiable *internal alignment embeddings* (IAE). Unlike external reward shaping or post-hoc safety constraints, IAE are learned latent variables that predict externalized harm through counterfactual reasoning and modulate policy gradients toward harm reduction via attention gating and graph diffusion.

We formalize the ESAI framework through four integrated mechanisms: (1) differentiable counterfactual alignment penalties computed via softmax reference distributions, (2) IAE-weighted attention biasing perceptual salience toward alignment-relevant features, (3) Hebbian affect-memory coupling supporting temporal credit assignment, and (4) similarity-weighted graph diffusion with bias-mitigation controls. We derive conditions for bounded internal embeddings under Lipschitz constraints and spectral radius bounds, analyze computational complexity as $O(N|\mathcal{A}|k^2 + N|\mathcal{A}|kd)$ dominated by counterfactual forecasting for N agents with k -dimensional embeddings and action space size $|\mathcal{A}|$, and discuss theoretical properties including contraction dynamics and fairness-performance tradeoffs.

This work positions ESAI as a conceptual contribution to differentiable alignment mechanisms in multi-agent systems. We identify open theoretical questions regarding convergence guarantees, optimal embedding dimensionality, and extension to high-dimensional state spaces. Empirical validation remains future work.

CCS Concepts

• Computing methodologies → Reinforcement learning; Multi-agent systems.

Keywords

embedded safety-aligned intelligence, internal alignment embeddings, multi-agent reinforcement learning, counterfactual reasoning, harm reduction, attention gating, graph diffusion, safe reinforcement learning

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1 Introduction

Contemporary Multi-Agent Reinforcement Learning (MARL) optimizes explicit task objectives but typically lacks internal differentiable regulators that encourage prosocial behavior and stable coordination under distribution shift. Standard approaches to alignment—such as reward shaping [16], constrained optimization [1], or inverse reinforcement learning [8]—rely on external supervision

signals that are either hand-designed, require human preference data, or operate as non-differentiable constraints decoupled from policy learning dynamics.

We investigate whether agents can instead learn an *Internal Alignment Embedding* (IAE): a differentiable latent variable that tracks predicted externalized harm and shapes policy gradients toward harm reduction through three key properties:

- (1) **Predictive:** IAE forecasts alignment-relevant outcomes via counterfactual reasoning over candidate actions.
- (2) **Regulatory:** IAE modulates perception and action selection through attention gating and gradient coupling.
- (3) **Distributed:** IAE propagates across agent neighborhoods via graph diffusion with controllable similarity weighting.

This paper presents *Embedded Safety-Aligned Intelligence* (ESAI), a theoretical framework formalizing these principles. ESAI differs from existing alignment approaches in three fundamental ways: (1) alignment constraints are *embedded* in learned internal representations rather than imposed externally, (2) harm prediction is *differentiable* and jointly trained with policy parameters, and (3) multi-agent coordination emerges from *graph diffusion* of alignment signals rather than centralized control or explicit communication protocols.

Contributions. We formalize ESAI as a conceptual framework through:

- A differentiable counterfactual alignment penalty enabling gradient-based harm forecasting without discrete argmin operations (Section ??).
- IAE-weighted attention mechanisms biasing perceptual salience toward alignment-relevant features (Section 3.5).
- Hebbian affect-memory coupling with differentiable read operations supporting temporal credit assignment (Section 3.6).
- Locally damped graph diffusion with similarity-weighted propagation and bias-mitigation controls (Section 3.7).
- Stability conditions for bounded internal embeddings under Lipschitz constraints and spectral radius bounds (Section 4.1).

Scope and positioning. ESAI is a *conceptual framework*, not a validated algorithm. We present one possible architectural instantiation but emphasize that alternative implementations may satisfy ESAI principles. Key open questions include optimal embedding dimensionality, convergence guarantees under stochastic dynamics, and scalability to high-dimensional observation spaces. **Empirical validation across diverse environments remains essential future work** to assess whether ESAI’s theoretical advantages translate to practical coordination improvements.

We note that IAE is a computational abstraction—*not* a claim about subjective emotion or consciousness. All theoretical analysis assumes availability of domain-specific harm metrics; implications of this assumption are discussed in Section 5.3.

2 Related Work

Alignment and safety in MARL. Cooperative inverse reinforcement learning [8] infers alignment objectives from human demonstrations but requires extensive preference data and does not maintain differentiable internal alignment states. Deep reinforcement learning from human preferences [5] learns reward models from comparisons but operates on scalar rewards rather than internal representations that can gate perception or modulate credit assignment.

Constrained Policy Optimization (CPO) [1] enforces safety via Lagrangian constraints but lacks differentiable internal alignment states that persist across timesteps. Safe reinforcement learning via shielding [3] uses formal methods to prevent unsafe actions but operates as an external constraint layer rather than an embedded internal mechanism. Recent work on shielded MARL [4] introduces probabilistic logic shields that reduce safety violations through external symbolic constraints operating separately from policy learning.

Intrinsic inequity aversion [10] demonstrates that internal social preferences improve cooperation in multi-agent settings. While this shows that intrinsic motivations can enhance coordination, the preference structure is hand-designed rather than learned from predicted outcomes. Multi-objective reinforcement learning [9] optimizes Pareto frontiers over competing objectives but does not integrate alignment-specific internal representations.

ESAI contributes a differentiable framework where alignment emerges from learned internal embeddings that predict harm via counterfactual reasoning and modulate policy gradients through attention gating, rather than relying on external rewards, hand-designed preferences, or non-differentiable constraints.

Counterfactual reasoning in MARL. Counterfactual Multi-Agent Policy Gradients [6] computes counterfactual baselines by marginalizing single-agent actions for centralized credit assignment. These counterfactuals reduce variance in gradient estimates but do not predict or prevent externalized harm. Prediction-assisted counterfactual methods [18] use counterfactual predictions as auxiliary losses to assist value factorization, demonstrating that learned counterfactual models can improve representational quality, but target credit assignment rather than alignment.

Work on measuring kindness in MARL [2] is conceptually closest to ESAI in exploring intrinsic prosocial mechanisms. However, it focuses on defining and measuring kindness metrics rather than learning differentiable internal alignment states that forecast harm through counterfactual reasoning and gradient flow.

ESAI extends counterfactual reasoning from credit assignment to differentiable alignment penalties, training IAE to forecast harm and penalizing deviations from harm-minimizing reference policies via softmin distributions that preserve gradient flow.

Graph networks and attention in MARL. Graph neural networks enable structured communication in MARL by modeling agents as nodes with learned edge weights [12]. Attention mechanisms compute dynamic edge weights so agents learn which peers are relevant in each state. Actor-attention-critic methods [11] show that attention over other agents improves coordination. Graph attention for multi-agent game abstraction [13] and duplex dueling

with attention for value factorization [17] demonstrate improved performance through learned relational representations. Neural attention additive models [14] show that attention provides both performance and interpretability.

However, existing work does not integrate graph mechanisms with internal alignment states that track predicted harm.

ESAI integrates graph diffusion with IAE dynamics, enabling similarity-weighted propagation of alignment-relevant information with explicit bias-mitigation controls.

Memory and plasticity. Differentiable memory systems such as Neural Turing Machines [7] enable learned read/write operations for temporal credit assignment. Differentiable Hebbian plasticity [15] demonstrates that backpropagation can train associative learning rules that adapt network weights during deployment.

However, no existing work couples Hebbian learning with alignment-specific internal states or uses Hebbian traces to support counterfactual forecasting of harm.

ESAI couples Hebbian traces to IAE dynamics via differentiable read operations, enabling alignment-aware memory updates that support counterfactual forecasting.

Summary of gaps addressed. While individual components exist in isolation—counterfactual reasoning for credit assignment, graph diffusion for communication, attention for coordination, external shields, and differentiable memory—**no existing work unifies differentiable internal alignment embeddings with counterfactual harm prediction, graph-based propagation, Hebbian memory coupling, and bias-mitigation controls in a single end-to-end trainable architecture.** ESAI provides this unification as a theoretical framework.

3 ESAI Framework

This section formalizes the ESAI framework, presenting definitions, architectural components, and mathematical formulations. Figure 1 illustrates the complete architecture.

3.1 Definitions

DEFINITION 1 (INTERNAL ALIGNMENT EMBEDDING). *An IAE is a differentiable latent variable $E_t \in \mathbb{R}^k$ maintained by each agent satisfying:*

- (1) **Predictive correspondence:** E_t correlates with predicted externalized harm under learned dynamics.
- (2) **Gradient coupling:** E_t influences policy gradients via differentiable transformations of rewards or observations.
- (3) **Temporal persistence:** E_t evolves through learned update rules preserving information across timesteps.

DEFINITION 2 (EMBEDDED SAFETY-ALIGNED INTELLIGENCE). *A multi-agent learning system exhibits ESAI if:*

- (1) *Each agent maintains an IAE satisfying Definition 1.*
- (2) *Policy learning incorporates a differentiable alignment objective penalizing predicted harm.*
- (3) *IAE dynamics are supervised to forecast alignment-relevant outcomes.*
- (4) *The system supports distributed coordination through IAE propagation across agent neighborhoods.*

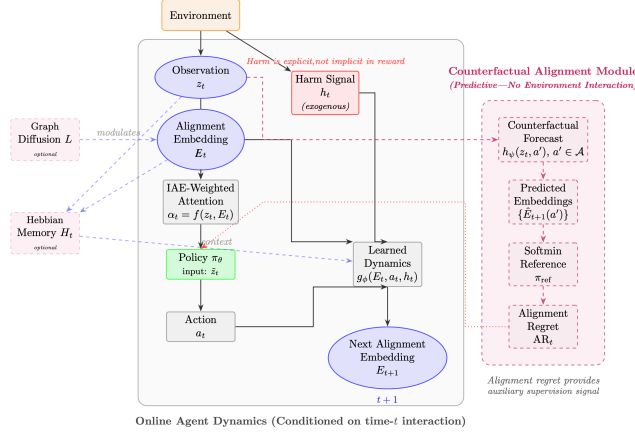


Figure 1: ESAI architecture. The internal alignment embedding E_t is updated via learned dynamics g_ϕ , graph diffusion L , and Hebbian memory H_t . Counterfactual forecasts generate alignment penalties AR_t that shape policy gradients. IAE-weighted attention α_t modulates perceptual input z_t . Dotted lines indicate gradient flow; solid lines denote forward computation.

ESAI systems differ from external alignment mechanisms in that alignment pressure arises from *internal learned representations* rather than external supervisory signals, enabling gradient-based adaptation, perceptual salience modulation, and distributed coordination without centralized oversight.

3.2 Task and Alignment Objectives

Let $s_t \in \mathcal{S}$ denote environment state, $a_t \in \mathcal{A}$ agent action, and r_t^{ext} extrinsic reward. The standard reinforcement learning objective is:

$$J_{\text{task}}(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r_t^{\text{ext}} \right]. \quad (1)$$

ASSUMPTION 1 (HARM OBSERVABILITY). *There exists a measurable harm signal $h_t : \mathcal{S} \times \mathcal{A}^N \times \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$ that is exogenous (defined independently of policy parameters), observable from transitions, and bounded. ESAI does not learn what constitutes harm—only how to predict and avoid externally specified harm.*

Each agent $i \in \{1, \dots, N\}$ maintains IAE $E_{i,t} \in \mathbb{R}^k$. The alignment objective encourages low-norm embeddings:

$$J_{\text{align}}(\theta) = -\mathbb{E}_{\pi_\theta} \left[\sum_t \gamma^t \sum_{i=1}^N \|E_{i,t}\|_2 \right], \quad (2)$$

encoding the principle that lower IAE magnitude corresponds to lower predicted harm.

3.3 IAE Dynamics

The embedding for agent i evolves via:

$$E_{i,t+1} = \gamma_E E_{i,t} + g_\phi(z_{i,t}, a_{i,t}, r_{i,t}^{\text{ext}}) - \alpha \sum_{j \in \mathcal{N}(i)} L_{ij} E_{j,t}, \quad (3)$$

where $\gamma_E \in [0, 1)$ controls temporal persistence, g_ϕ is a learned update function (e.g., a Multi-Layer Perceptron, MLP), $z_{i,t}$ is agent

i 's observation, $\mathcal{N}(i)$ is agent i 's neighborhood, L_{ij} are entries of the normalized graph Laplacian, and $\alpha \geq 0$ controls diffusion strength.

The diffusion term propagates alignment-relevant information across agent neighborhoods, enabling decentralized coordination: agents in similar states with high graph connectivity develop correlated IAE dynamics.

3.4 Differentiable Counterfactual Alignment Penalty

For each candidate action $a \in \mathcal{A}$, agent i forecasts the next-step IAE:

$$\widehat{E}_{i,t+1}^{(a)} = h_\psi(z_{i,t}, a, r_{i,t}^{\text{ext}}, \text{read}(H_{i,t})), \quad (4)$$

where h_ψ is a learned forecast network and $\text{read}(H_{i,t})$ provides Hebbian memory context (Section 3.6).

To avoid non-differentiable arg min, we define a softmin reference distribution with temperature τ :

$$\pi_{\text{ref}}(a | s_t) = \frac{\exp(-\|\widehat{E}_{i,t+1}^{(a)}\|_2 / \tau)}{\sum_{a'} \exp(-\|\widehat{E}_{i,t+1}^{(a')}\|_2 / \tau)}. \quad (5)$$

The expected reference embedding is $\widehat{E}_{i,t+1}^{\text{ref}} = \sum_a \pi_{\text{ref}}(a | s_t) \widehat{E}_{i,t+1}^{(a)}$. The differentiable alignment regret penalizes deviations from the harm-minimizing reference:

$$AR_{i,t} = \|E_{i,t+1} - \widehat{E}_{i,t+1}^{\text{ref}}\|_2^2 + \kappa \cdot \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \|E_{j,t}\|_2, \quad (6)$$

where the second term regularizes neighbor embeddings using lagged values $E_{j,t}$ for causal consistency.

Predictor stability. To prevent predictor-policy collusion, we maintain an Exponential Moving Average (EMA) target network $\psi_{\text{target}} \leftarrow \tau_{\text{ema}} \psi_{\text{target}} + (1 - \tau_{\text{ema}}) \psi$, with counterfactual forecasts computed using $h_{\psi_{\text{target}}}$ to stabilize gradient flow.

3.5 IAE-Weighted Attention

Attention weights bias perceptual salience based on current IAE:

$$\alpha_{i,t} = \text{softmax}(W_a E_{i,t} + b_a) \in \mathbb{R}^d, \quad \tilde{z}_{i,t} = \alpha_{i,t} \odot z_{i,t}, \quad (7)$$

where $W_a \in \mathbb{R}^{d \times k}$ is a projection matrix and $\odot$ denotes element-wise product. In sparse-harm environments, this mechanism enables learned salience modulation toward alignment-relevant features.

3.6 Hebbian Affect-Memory Coupling

A Hebbian memory matrix $H_{i,t} \in \mathbb{R}^{k \times d}$ updates via outer-product learning:

$$H_{i,t+1} = (1 - \delta_H) H_{i,t} + \eta_H (E_{i,t} \otimes \tilde{z}_{i,t}), \quad (8)$$

where $\delta_H > 0$ ensures decay and η_H is the learning rate. The trace supports counterfactual forecasting via differentiable read: $\text{read}(H_{i,t}) = W_r \text{vec}(H_{i,t})$. Hebbian traces encode historical co-activations, providing context for harm prediction and temporal credit assignment.

3.7 Graph Diffusion with Bias Mitigation

The graph Laplacian L in Eq. (3) is row-normalized with spectral radius $\rho(L) \leq 2$. Diffusion weights are modulated by cosine similarity of learned identity embeddings ϕ_I :

$$\beta_{ij} = \max(0, \cos(\phi_i, \phi_j)). \quad (9)$$

To mitigate emergent in-group favoritism, we introduce a similarity-suppression regularizer:

$$L_{\text{bias}} = \lambda_{\text{bias}} \|\tilde{A} \odot S\|_F^2, \quad (10)$$

where $S_{ij} = \beta_{ij}$ and $\tilde{A}$ is the weighted adjacency. This provides tunable fairness-performance tradeoffs.

3.8 Policy Optimization

Extrinsic rewards are transformed with alignment penalties:

$$r'_{i,t} = r_{i,t}^{\text{ext}} - \lambda_{\text{reg}} \text{AR}_{i,t}. \quad (11)$$

Advantages $A_{i,t}$ are computed using Generalized Advantage Estimation (GAE) on $r'_{i,t}$. The policy loss follows Proximal Policy Optimization with Clipping (PPO-Clip):

$$L_{\pi} = \mathbb{E}_t \left[\min(\rho_t A_{i,t}, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) A_{i,t}) \right], \quad (12)$$

where $\rho_t = \pi_{\theta}(a_{i,t} | s_t) / \pi_{\theta_{\text{old}}}(a_{i,t} | s_t)$. ESAI principles are compatible with alternative policy gradient methods.

The full objective combines policy loss, entropy regularization, and auxiliary penalties:

$$L(\theta) = \mathbb{E}_t \left[L_{\pi} - \beta H(\pi_{\theta}) + \lambda_H \|H_{i,t}\|_2^2 + \lambda_D \|L\|_F^2 + L_{\text{bias}} \right]. \quad (13)$$

4 Theoretical Properties

We establish stability conditions ensuring bounded IAE dynamics and analyze computational complexity.

4.1 Stability Analysis

PROPOSITION 1 (BOUNDED IAE UNDER CONTRACTION). *Consider the IAE update in Eq. (3). Assume:*

- (1) g_{ϕ} is L_g -Lipschitz continuous with bounded output at origin.
- (2) Inputs are bounded: $\|z_{i,t}\|_2 \leq C_z$, $\|a_{i,t}\|_2 \leq C_a$, $|r_{i,t}| \leq C_r$ for all i, t .
- (3) The spectral condition holds: $\|\gamma_E I - \alpha L\|_2 + L_g < 1$.

Then $\sup_t \|E_{i,t}\|_2 < \infty$ for all agents i and trajectories.

PROOF SKETCH. By Lipschitz continuity and input boundedness, $\|g_{\phi}(z_{i,t}, a_{i,t}, r_{i,t})\|_2 \leq K$ for some constant K . Taking norms in Eq. (3):

$$\|E_{i,t+1}\|_2 \leq \|\gamma_E I - \alpha L\|_2 \max_j \|E_{j,t}\|_2 + K.$$

With $\rho = \|\gamma_E I - \alpha L\|_2 < 1 - L_g$, iteration yields $\max_i \|E_{i,t}\|_2 \leq \rho^t \max_i \|E_{i,0}\|_2 + K/(1 - \rho) < \infty$. The spectral condition can be enforced via spectral normalization of L . $\square$

REMARK 1. Proposition 1 guarantees bounded IAE, preventing numerical instability. However, bounded embeddings are necessary but not sufficient for aligned behavior. Convergence to socially optimal equilibria remains an open theoretical problem.

Table 1: Per-agent computational complexity. Here d is observation dimension, k is IAE dimension, and $|\mathcal{A}|$ is action space size.

Component	Complexity
IAE dynamics (Eq. 3)	$O(k^2 + kd + \mathcal{N} k)$
Counterfactual forecasts (Eq. 4)	$O(\mathcal{A} (k^2 + kd))$
Softmin reference (Eq. 5)	$O(\mathcal{A} k)$
Attention gating (Eq. 7)	$O(kd)$
Hebbian update (Eq. 8)	$O(kd)$
Total per agent	$O(\mathcal{A} k^2 + \mathcal{A} kd)$

PROPOSITION 2 (HEBBIAN TRACE STABILITY). *If the state-IAE joint distribution admits bounded second moments and the design constraint $\delta_H > \eta_H \sqrt{\mathbb{E}[\|E_{i,t}\|_2^2] \mathbb{E}[\|z_{i,t}\|_2^2]}$ holds, then the Hebbian trace converges in mean-square sense to a bounded fixed point.*

PROOF SKETCH. Taking expectations in Eq. (8) and applying Cauchy-Schwarz, the update forms a contraction when the decay rate δ_H exceeds the expected outer-product magnitude scaled by η_H . The fixed point satisfies $\mathbb{E}[\|H^*\|_F] = \eta_H \mathbb{E}[\|E_{i,t}\|_2 \|z_{i,t}\|_2] / \delta_H < \infty$. $\square$

4.2 Computational Complexity

Table 1 summarizes per-agent computational costs. For N agents with bounded-degree graphs ($|\mathcal{N}(i)| = O(1)$), total complexity per timestep is $O(N|\mathcal{A}|k^2 + N|\mathcal{A}|kd)$, dominated by counterfactual forecasting. This can be mitigated via top- K action sampling, reducing effective action space from $|\mathcal{A}|$ to $K \ll |\mathcal{A}|$.

Overhead relative to standard methods. Compared to vanilla PPO with complexity $O(d^2)$ per agent, ESAI introduces overhead factor approximately $(|\mathcal{A}|k^2 + |\mathcal{A}|kd)/d^2$. For typical values ($d = 64$, $k = 32$, $|\mathcal{A}| = 6$), this yields $\sim 4.5\times$ overhead—manageable for moderate-scale problems but motivating future work on efficient approximations.

5 Discussion

5.1 Potential Advantages of Embedded Alignment

ESAI’s theoretical design suggests several potential advantages over external alignment mechanisms:

Gradient-based harm forecasting. Unlike discrete safety constraints or non-differentiable shields, IAE enables continuous gradient flow from predicted harm to policy parameters. This could enable on-line adaptation to novel harm types without manual constraint redesign.

Perceptual salience modulation. IAE-weighted attention (Eq. 7) provides a mechanism for learned salience toward alignment-relevant features. In sparse-harm environments where harmful states are rare, this could improve sample efficiency compared to uniform attention.

Decentralized coordination. Graph diffusion of IAE enables distributed alignment pressure: agents in connected neighborhoods

develop correlated harm predictions without centralized oversight. This could improve scalability compared to centralized critics while maintaining coordination benefits.

Interpretable bias controls. The similarity-suppression regularizer (Eq. 10) provides an explicit tuning parameter for fairness-performance tradeoffs, enabling transparent auditing compared to black-box reward shaping.

5.2 Comparison to Alternative Paradigms

Versus reward shaping. Potential-based shaping [16] provides theoretical guarantees (policy invariance) that ESAI lacks. However, ESAI’s learned dynamics could adapt to non-stationary harm structures where hand-designed potentials fail. This tradeoff between guarantees and adaptive capacity requires empirical investigation.

Versus constrained optimization. CPO [1] enforces hard safety constraints via trust regions. ESAI trades hard guarantees for soft, differentiable penalties enabling gradient-based learning. In high-dimensional action spaces where constraint satisfaction is expensive, ESAI’s continuous relaxation may offer computational advantages at the cost of occasional constraint violations.

Versus multi-objective reinforcement learning. Multi-objective methods [9] optimize Pareto frontiers over competing objectives. ESAI implicitly performs multi-objective optimization via λ_{reg} but adds internal dynamics (attention, diffusion) unavailable to standard scalarization. Whether this complexity provides practical benefits is an empirical question.

5.3 Limitations and Assumptions

Dependence on harm specification. ESAI assumes availability of domain-specific harm metrics (Assumption 1). The framework learns to *predict and avoid* harm but does not learn *what constitutes harm*—this normative content must be externally specified. Defining harm is inherently a normative choice encoding cultural values that may not generalize across contexts.

Lack of empirical validation. This paper presents a theoretical framework without experimental validation. Claims about advantages are conjectural pending empirical investigation. Key empirical questions include: (1) Does ESAI improve cooperation compared to baselines? (2) Do IAE dynamics exhibit interpretable structure? (3) How does performance scale with embedding dimension, agent count, and connectivity?

Theoretical gaps. Our stability analysis (Propositions 1–2) provides sufficient conditions for bounded dynamics but does not address convergence to socially optimal equilibria, sample complexity bounds, or robustness guarantees under distribution shift.

Architectural complexity. ESAI integrates four mechanisms, each introducing hyperparameters. Whether simpler architectures achieve comparable alignment with fewer degrees of freedom is unknown. The principle of parsimony suggests validating each component’s necessity empirically.

5.4 Broader Impact and Ethical Considerations

Potential benefits. If empirically validated, ESAI could contribute to safer exploration in cooperative robotics, reduced need for extensive human preference labeling, and interpretable bias-detection controls.

Potential risks. Deployment of ESAI-based systems could create risks including: (1) *Bias amplification*: similarity-weighted diffusion can encode in-group favoritism if not carefully regularized; (2) *Normative lock-in*: harm specifications entrench values of system designers, risking marginalization of minority perspectives; (3) *Dual-use potential*: differentiable coordination mechanisms could be repurposed for adversarial applications.

Governance requirements. Before real-world deployment, ESAI systems require domain-specific safety audits, diverse harm specification sources, continuous bias monitoring, adversarial stress testing, and human-in-the-loop oversight for high-stakes decisions.

We emphasize that IAE is a computational abstraction—not a model of subjective consciousness or human emotion.

5.5 Open Theoretical Questions

Several fundamental questions remain unresolved and represent directions for future research:

- **Optimal embedding dimensionality:** What is the minimal k required to represent alignment-relevant structure? Information-theoretic bounds could formalize this relationship.
- **Convergence guarantees:** Under what conditions do ESAI policies converge to socially optimal Nash equilibria rather than selfish or collusive outcomes?
- **Robustness to distribution shift:** How do IAE dynamics generalize when deployed in environments with different harm structures than training?
- **Interpretability:** Can we derive conditions under which IAE factors into semantically meaningful subspaces via disentanglement objectives?

6 Conclusion

We introduced ESAI, a theoretical framework for MARL that embeds alignment constraints in learned internal representations. Through IAE supervised via counterfactual harm prediction, ESAI provides a conceptual alternative to external reward shaping and non-differentiable safety constraints.

Key contributions. The framework integrates four mechanisms: (1) differentiable counterfactual alignment penalties enabling gradient-based harm forecasting, (2) IAE-weighted attention biasing perceptual salience, (3) Hebbian affect-memory coupling supporting temporal credit assignment, and (4) similarity-weighted graph diffusion with bias-mitigation controls. We derived stability conditions for bounded embeddings (Propositions 1–2) and analyzed computational complexity as $O(N|\mathcal{A}|k^2 + N|\mathcal{A}|kd)$.

Limitations. ESAI is a conceptual framework without empirical validation. Claims about advantages remain conjectural, and theoretical analysis provides bounded dynamics but not convergence guarantees to socially optimal equilibria.

Future work. Critical next steps include: (1) empirical validation across diverse MARL benchmarks with statistical comparisons to reward shaping, CPO, and multi-objective methods; (2) convergence analysis and sample complexity bounds; (3) interpretability methods grounding IAE semantics in domain harm metrics; and (4) robustness characterization under distribution shift and adversarial perturbations.

ESAI demonstrates that embedding alignment in internal learned representations is a theoretically coherent paradigm for safe multi-agent coordination. Whether this translates to practical advantages over simpler external mechanisms remains an important empirical question—one we hope this framework motivates the community to investigate.

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